

JACOB ROME

651-447-9428 | jrome8602@gmail.com | Jake-Rome.com | St. Paul, MN | US and EU Citizen

EDUCATION

Georgia Institute of Technology

Bachelor of Science in Aerospace Engineering; Cumulative GPA: 3.93/4.00

Atlanta, GA

Aug. 2024 – Dec 2027

University of Minnesota

Dual Enrollment; Cumulative GPA: 4.00/4.00

Minneapolis, MN

Sept. 2023 – May. 2024

EXPERIENCE

Georgia Tech Aircraft Prototyping Laboratory (APL)

Undergraduate Research Assistant

Jun. 2026 – Current

Atlanta, GA

- Designed and CNC-manufactured carbon fiber airframe components, performing FEA to validate structural performance
- Manufactured aluminum molds for in-house propeller production, benchmarking thrust against off-the-shelf propellers

NASA RAVEN Project

Test Engineer

Summer 2026

Atlanta, GA

- Optimized a MATLAB script to test liquid cooling performance for motors on a 500V, 25-40A eVTOL surrogate vehicle
- Reduced required pump voltage 62%(24V→9V, 4.6→1.6 LPM), enabling a smaller pump/battery and lighter aircraft

Drone, Radio Control, Experimental Aircraft Club (DRXC)

Chief Engineer, Winner of the 2026 VFS DBVF Competition

Aug. 2025 – Current

Atlanta, GA

- Led system-level UAV design as Chief Engineer, coordinating 6 subteams through weekly technical design reviews
- Reduced airframe mass by 20% using SolidWorks topology optimization, FEA, and lightweight structural redesign
- Designed a vision-guided linear rail mechanism to autonomously capture and secure a 3-lb payload during flight

UpperStory

Research and Development Intern

Summer 2024 & Summer 2025

St. Paul, MN

- Designed from-scratch charger PCB and battery interface PCB enabling modular dual-battery power & charging system
- Validated battery terminal interface under 800 mA continuous load to ensure reliable electrical contact design
- Added controller functionality by redesigning PCB inputs with a button matrix and integrating a bumper system
- Engineered a 4WD wind-up car drivetrain, improving uphill performance using design iteration and resin 3D printing

NASA L'SPACE NPWEE

Project Manager and Primary Reviewer

Jan. 2025 – May. 2025

Atlanta, GA

- Led team developing NASA NPWEE proposal on spacecraft debris protection, managing risk, budget, and schedule
- Developed NASA proposal using requirements-driven design informed by feedback from NASA engineers and mentors
- Chaired formal proposal review, evaluating another team's design against NASA NPWEE scoring criteria and rubric

Yellow Jacket Space Program

Darcy Rocket Structures Team

Jan. 2025 – Aug. 2025

Atlanta, GA

- Engineered and designed a mold for the Darcy nose cone, which was optimized for manufacturing constraints
- 3D printed the molds with ABS and used acetone surface treatment to create a smooth finish for the layup process

FIRST Tech Challenge Robotics

Team Captain

Sept. 2019 – Feb. 2024

Woodbury, MN

- Led the team to appear in 3 consecutive state championship tournaments, winning awards in each appearance
- Directed design and manufacturing sub-teams, introducing new systems to produce 160+ 3D printed parts
- Received the Dean's List Finalist award and represented Minnesota at the World Championship tournament

PROJECTS

- 3D Printed Clock:** Engineered and assembled a fully 3D printed clock with a torsional energy system and escapement mechanism, modeling components in CAD and iterating through prototypes to achieve reliable timekeeping
- Tensile Test Machine:** Designed and built a tensile test machine using a load cell and Arduino to measure stress-strain behavior of multiple materials, identifying of material properties through data acquisition and analysis
- Rubik's Cube Solver:** Developed a Rubik's Cube solving robot using a color sensor for initial state detection, stepper motors for precise actuation, and algorithmic solution planning to autonomously solve the cube
- Portfolio Website:** Designed and deployed an interactive portfolio website using React and JavaScript to showcase engineering projects, technical experiences, and CAD models in a structured, user friendly format

SKILLS

Technical: CAD, FEA, Electrical Schematics, PCB Design, GD&T, Fluid Dynamics, Control Systems

Tools: 3D Printing (FDM, Resin), Laser Cutter, CNC Mill, Microcontrollers, PCB Assembly, Transducers, Soldering

Programming: MATLAB, Simulink C/C++ , React, JavaScript, HTML

Software: Fusion 360, SolidWorks, Inventor, NX, ArduPilot, ANSYS, Arduino IDE, VS Code